FAST — NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
FAST School of Computing · Karachi Campus
AL-2002 Artificial Intelligence (Laboratory) · Fall 2026

PROJECT PROPOSAL

Smart Farming Predictor: AI-Powered Crop Recommendation

Group Information

Name	Roll Number	Section	Role in the project
Ali Noor	24I-2073	BCY-5A	Leader; coordination and integration
Gotam Lal	24K-2047	BCY-5A	AI implementation and evaluation

Project field	Details
Project Idea	Own idea
Category	Part B — Applied / Data
GitHub Repository URL	To be added
Date of Submission	21 October 2026

1. Project Title and Summary

Smart Farming Predictor is an AI-based application that recommends a suitable crop from soil and environmental conditions. It accepts nitrogen, phosphorus, and potassium levels, temperature, humidity, soil pH, and rainfall, then compares these inputs with crop requirements. The project combines A* search and a genetic algorithm with a Random Forest classifier used as a baseline for comparison. The system is intended to support farmers and agricultural learners in making more informed crop-selection decisions. It will display the recommended crop and comparison results so users can understand how the recommendation was produced.

2. Problem Statement and Motivation

Crop selection depends on several interacting conditions, including soil nutrients, pH, temperature, humidity, and rainfall. Farmers may rely on experience or general assumptions when they do not have access to soil analysis or agricultural advice. Choosing a crop that does not fit local conditions can reduce productivity and waste water, fertilizer, time, and money. Making a recommendation manually is difficult because each crop has a different range of suitable conditions and the input features use different scales. The project will use the Kaggle Crop Recommendation Dataset, described in the source proposal as containing 2,200 records, seven input features, and 22 crop classes. After normalizing the numerical features, the system will represent crop choices as a search problem and compare the search-based output with a Random Forest classifier. A genetic algorithm will also be explored for evaluating candidate farming plans against the available data. The project is worthwhile because it demonstrates how AI methods can support more consistent, data-informed crop selection.

3. Problem Formulation

3.1 PEAS Description

Component	Description
Performance measure	Correctness of recommendations against labelled test data; classifier accuracy; search cost and runtime; genetic-algorithm fitness.
Environment	A local Python application using crop/soil feature records from the locally stored Kaggle dataset; user supplies environmental values.
Actuators	Display recommended crop, comparison predictions, and evaluation metrics.
Sensors	User-entered N, P, K, temperature, humidity, pH, and rainfall values; dataset records during training/testing.

3.2 Environment and Agent Type

• Observability: Fully observable for each recommendation; all seven input features are provided.

• Determinism: A* and Random Forest are deterministic for fixed settings; the genetic algorithm is stochastic.

• Change over time: Static during a single recommendation run.

• State and actions: Discrete crop candidates with continuous-valued input features.

• Agents: Single-agent decision-support application.

• Agent type: Goal-based / utility-based search with a learning-based baseline.

3.3 Problem Representation

Element	Representation
State representation	A normalized seven-feature vector representing soil nutrients and environmental conditions, plus candidate crop nodes in the search graph.
Actions / moves	Evaluate a candidate crop, move to a connected candidate in the crop graph, or generate/modify a candidate farming-plan chromosome.
Initial state	The user's seven input values after validation and feature normalization.
Goal or terminal test	Return a crop recommendation when the search reaches the selected best candidate; for GA, stop after 200 generations or the configured stopping condition.
Cost, utility, fitness or evaluation	A* uses normalized Euclidean distance to a crop's representative/ideal feature vector as its heuristic; GA fitness is evaluated against available historical/labelled data; Random Forest predictions and accuracy provide the baseline.

4. Proposed AI Approach

Technique	Role in the project	Own code or library
A* Search	Searches a weighted crop-candidate graph using normalized feature distance to guide the recommendation.	Own code
Genetic Algorithm	Searches candidate farming plans using tournament selection, single-point crossover (rate 0.8), and 200 generations.	Own code
Random Forest Classifier	Baseline classifier trained on the Crop Recommendation Dataset; compare predictions and test accuracy with the search-based approach.	scikit-learn/library

Baseline: Random Forest Classifier.

A* search is proposed to rank crop candidates using normalized distances between the user's conditions and crop feature profiles. Min-max normalization is important because nutrient values, temperature, humidity, pH, and rainfall have different numeric ranges. The genetic algorithm provides a second search-based approach: candidate plans are encoded as chromosomes, selected through tournament selection, and combined using single-point crossover at a rate of 0.8 for up to 200 generations. A Random Forest classifier will act as a baseline so the project can compare a conventional supervised-learning prediction with the search-based recommendations. Results will be evaluated on the same held-out examples and with fixed random seeds where applicable. The distance heuristic and fitness definition will be documented and tested; claims of optimality will only be made if the implemented graph costs and heuristic satisfy the required conditions.

4.1 Planned Class Design

• CropDataset: Attributes: dataset path, feature columns, target column. Methods: load_csv(), clean_data(), split_data().

• FeatureProcessor: Attributes: feature names, scaling parameters. Methods: validate_input(), fit_scaler(), transform().

• AStarCropRecommender: Attributes: crop graph, heuristic, open list. Methods: build_graph(), heuristic_cost(), recommend().

• GeneticFarmPlanner: Attributes: population size, crossover rate, generation limit. Methods: initialize_population(), fitness(), selection(), crossover(), evolve().

• RandomForestBaseline: Attributes: classifier, training data. Methods: train(), predict(), evaluate().

• RecommendationApp: Attributes: model instances, current user input, results. Methods: get_input(), run_recommendation(), display_results().

5. Scope

#	Feature	Why it is essential
1	Input and validation for seven soil/environment features	Prevents invalid or incomplete user input.
2	Load and preprocess the local Kaggle CSV	Makes the application runnable from a local dataset.
3	Implement normalized A* crop recommendation	Delivers the primary search-based AI technique.
4	Implement genetic algorithm with documented settings	Provides a second AI technique for comparison.
5	Train/evaluate Random Forest baseline and show results	Provides a reference point for comparing recommendations.

Out of scope

Live weather feeds, paid APIs, hardware sensors, mobile deployment, yield guarantees, and a production-grade farm-management platform are excluded. The system will provide data-based suggestions, not replace professional agronomic advice. The proposal does not yet specify a verified GitHub repository URL or a confirmed project-idea number; these should be added once available.

6. Data or Environment Source

Field	Details
Source name and URL	Kaggle Crop Recommendation Dataset — https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset
Type	Public dataset; stored and read locally as CSV.
Access requirements	No paid source planned; dataset download/account requirements should be checked before submission.
Approximate size	2,200 records; 7 input features; 22 crop classes (as stated in the source proposal).
Licence / terms of use	Check and follow the dataset's current Kaggle licence/terms before redistribution.
Offline fallback	Keep the downloaded CSV in the project data/ folder. Load it from disk at runtime and show a clear error if the file is missing; no network access should be required for normal execution.

7. Evaluation Plan

Evaluation item	Plan
What will be compared	A* recommendation, Genetic Algorithm output, and Random Forest prediction.
Metric(s)	Accuracy on held-out labelled records; confusion matrix; average prediction/search runtime; A* nodes expanded; GA best fitness by generation.
Baseline	Random Forest classifier.
Test instances / test set	Use a fixed train/test split of the 2,200-record dataset; keep the same held-out records for all comparable methods.
Number of runs and seed	At least 5 runs for stochastic GA experiments; record and fix random seeds. Use the same split and seeds for reproducibility.
Planned charts (at least three)	(1) Test accuracy by approach; (2) runtime by approach; (3) GA best fitness by generation; optionally (4) confusion matrix.

8. Division of Work

Member	Modules / components owned	Approximate share
Ali Noor — 24I-2073	Project coordination, dataset preparation, integration and final evaluation.	50%
Gotam Lal — 24K-2047	A* search, genetic algorithm, Random Forest baseline, and result visualizations.	50%

Roles and shares are proposed and should be confirmed by the group.

9. Milestones and Timeline

#	Demonstrable milestone	Target date	Owner
1	Dataset loading, input validation, and feature preprocessing demonstrably run.	Week 11 / date to confirm	Ali Noor
2	A* recommender returns a crop for test inputs and records search metrics.	Week 12 / date to confirm	Gotam Lal
3	Genetic algorithm and Random Forest baseline run on the same dataset split.	Week 13 / date to confirm	Ali Noor
4	Integrated application displays predictions and comparison results.	Week 14 / date to confirm	All members
5	Experiments complete, final integration, documentation and repository cleanup.	Week 15 / date to confirm	All members

10. Expected Outcome and Risks

A working submission will let a user enter seven soil and environmental values and receive a crop recommendation. The application will also show the A* result, genetic-algorithm result where applicable, and Random Forest baseline prediction, along with measured evaluation results. We expect the evaluation to reveal how recommendation quality and runtime differ between the approaches; no accuracy value is assumed before experiments are run. Two main risks are that the crop graph or heuristic may not represent crop

suitability correctly, and that the GA may produce unstable results across runs. To reduce these risks, the team will document graph construction, test representative cases, use fixed seeds, run repeated trials, and report limitations honestly.

References

• Atkar, A. (2020). Crop Recommendation Dataset. Kaggle. https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset

• Streamlit Documentation. https://docs.streamlit.io

• Russell, S. & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.). Pearson. A* Search, Chapter 3; Genetic Algorithms, Chapter 4.